Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative deadwood class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

ΔRMSE = RMSE(with M) − RMSE(without M)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPPsat (µmol m⁻² s⁻¹)	53% improved	59% improved	56% improved	56% improved
NEPmax (µmol m⁻² s⁻¹)	72% improved	69% improved	68% improved	69% improved
ETmax (mm d⁻¹)	62% improved	54% improved	57% improved	57% improved
WUE (g C mm⁻¹)	50% improved	54% improved	52% improved	50% improved

Relative deadwood
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 7.5 %, 37 sites)
- Mid (7.5–17.5 %, 42 sites)
- High (> 17.5 %, 22 sites)

Sites (sorted by ΔRMSE within each panel)